

ASTE-331a: Spacecraft Systems Engineering

Electrical Power Subsystem

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Electrical Power System (EPS) Overview

Power

Introduction

• Overview

• Examples

Power Generation

• Solar arrays

• RTGs

Energy Storage

• Batteries

• Other

Power Control

• Power regulation

• Power distribution

• Energy mgmt

Electrical Isolation &

Grounding

• Isolation

• Grounding

• EMI/EMC

Design Process

Case Studies

• Function

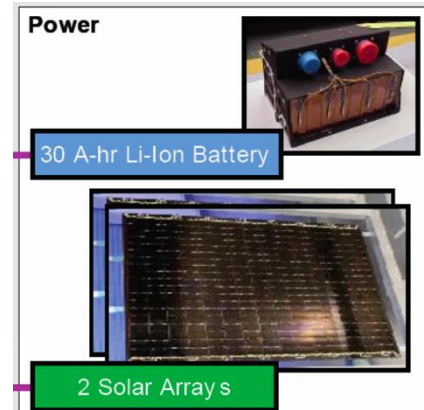
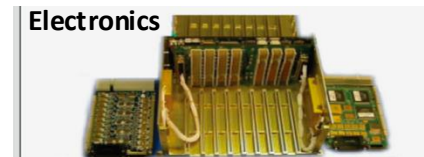
- Provide electrical power and distribution throughout the spacecraft
- Primary drivers are typically:
 - Sun-spacecraft distance
 - Power profiles during science and telecom operations
 - Launch and eclipses for battery sizing
 - Safe mode operation, including spacecraft attitude and power draw
 - Hardware redundancy

• Common Components

- Power generation
 - Solar arrays
 - Radioisotope thermoelectric generators (RTGs)
- Energy storage
 - Batteries, fuel cells
- Power control
 - Regulate array and battery voltage, distribute current from array and battery, charge/discharge battery
- Electrical isolation and grounding

• GRAIL Example

- Power electronics (interface cards, converters)
- 1 30 Ahr battery
- 2 solar arrays
- Total = ~30 kg



• Key Trades & Analyses

- Solar arrays vs. RTGs
- Battery type (e.g., primary vs. secondary)
- Solar array & battery sizing
- Power equipment list (PEL) and power analysis
- Heritage from prior systems

• Key Parameters

- System power loads, profiles, and margins
- Worst-case depth of discharge

EPS Overview

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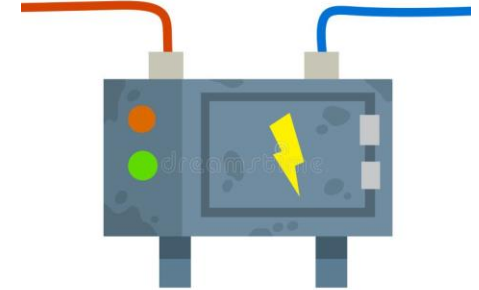
Power generation

- Solar power
- Nuclear power



Energy storage

- Batteries
- Fuel cells
- Flywheels
- Supercapacitors



Power control

- Power regulation
- Power distribution
- Energy management

EPS Overview

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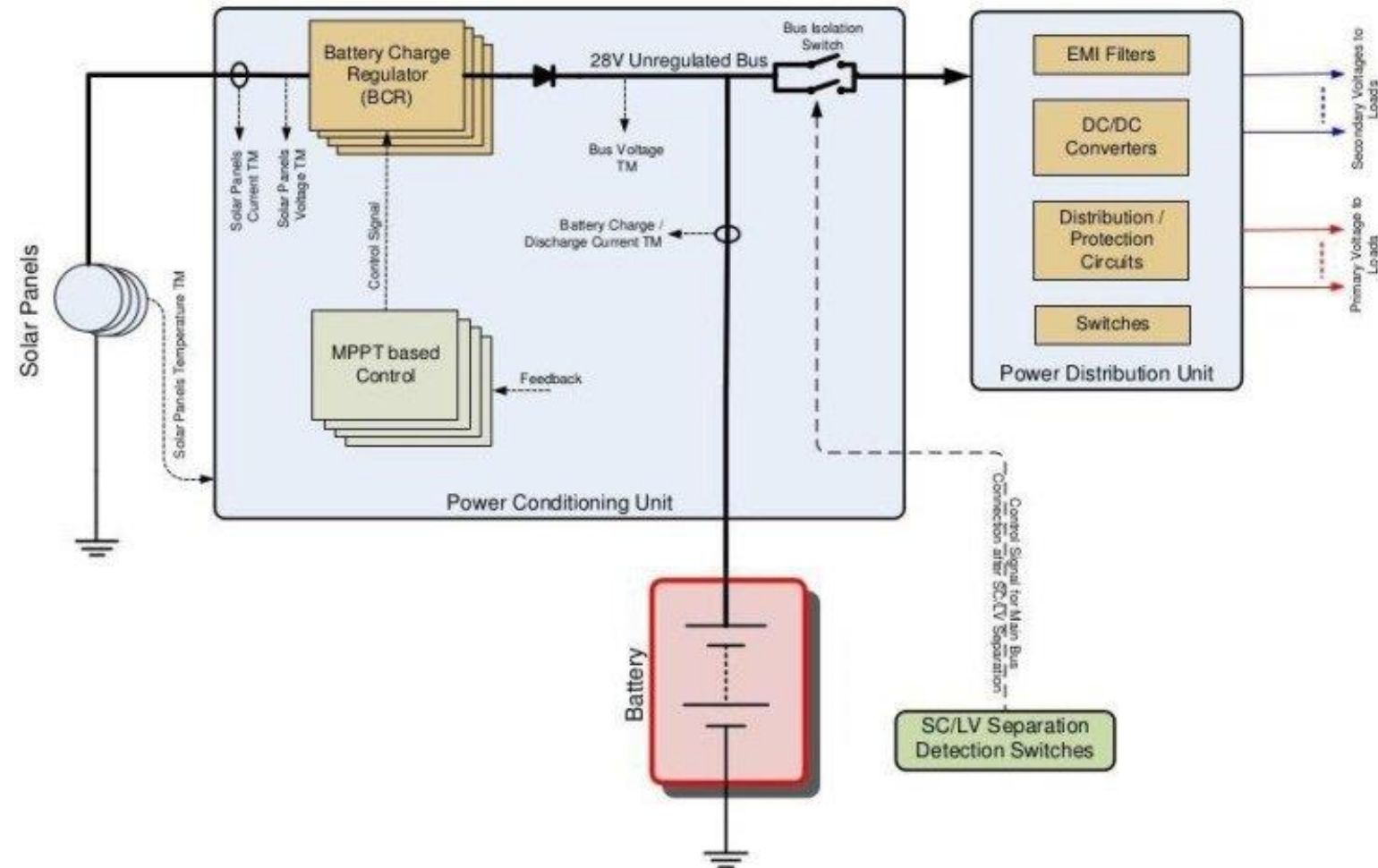
Electrical Isolation & Grounding

- Isolation
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Design Process

Case Studies

- **Typical components:**
 - Solar array
 - Power generation
 - Battery
 - Energy storage
 - Power conditioning unit (PCU)
 - Power regulation for solar array and battery
 - Power distribution unit (PDU)
 - Power conversion and switching for spacecraft devices



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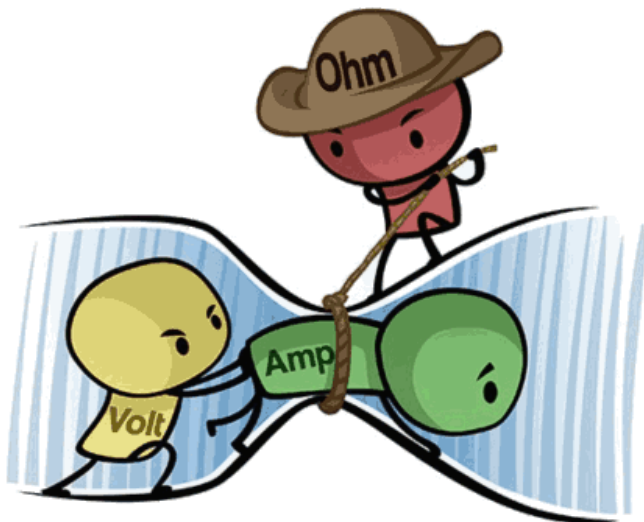
• EMI/EMC

Design Process

Case Studies

• Terminology reminders:

- **Voltage (Volts)** is like the pressure of water in a hose - it is the force that pushes electric charges through a wire
- **Current (Amperes)** is like the amount of water flowing through the hose - it measures how much electricity is moving through the wire
- **Resistance (Ohms)** is like sand stuck in the hose – it measures how much a material opposes the flow of electricity



Electricity is like a water hose

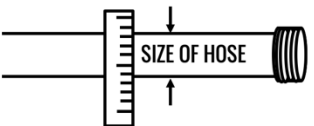
Voltage

Volts (V)



Current

Amps (A or I)



Resistance

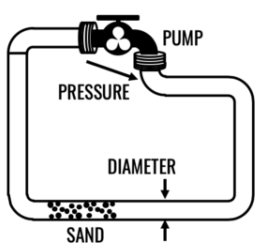
Ohms (R or Ω)



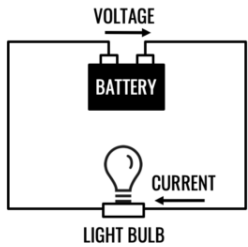
$$\text{Voltage} = \text{Current} \times \text{Resistance}$$

$$(V = I \times R)$$

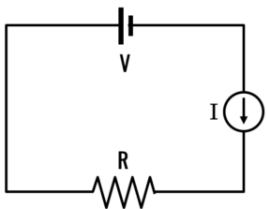
Water



Electricity



Circuit Diagram



Example: Solar Probe

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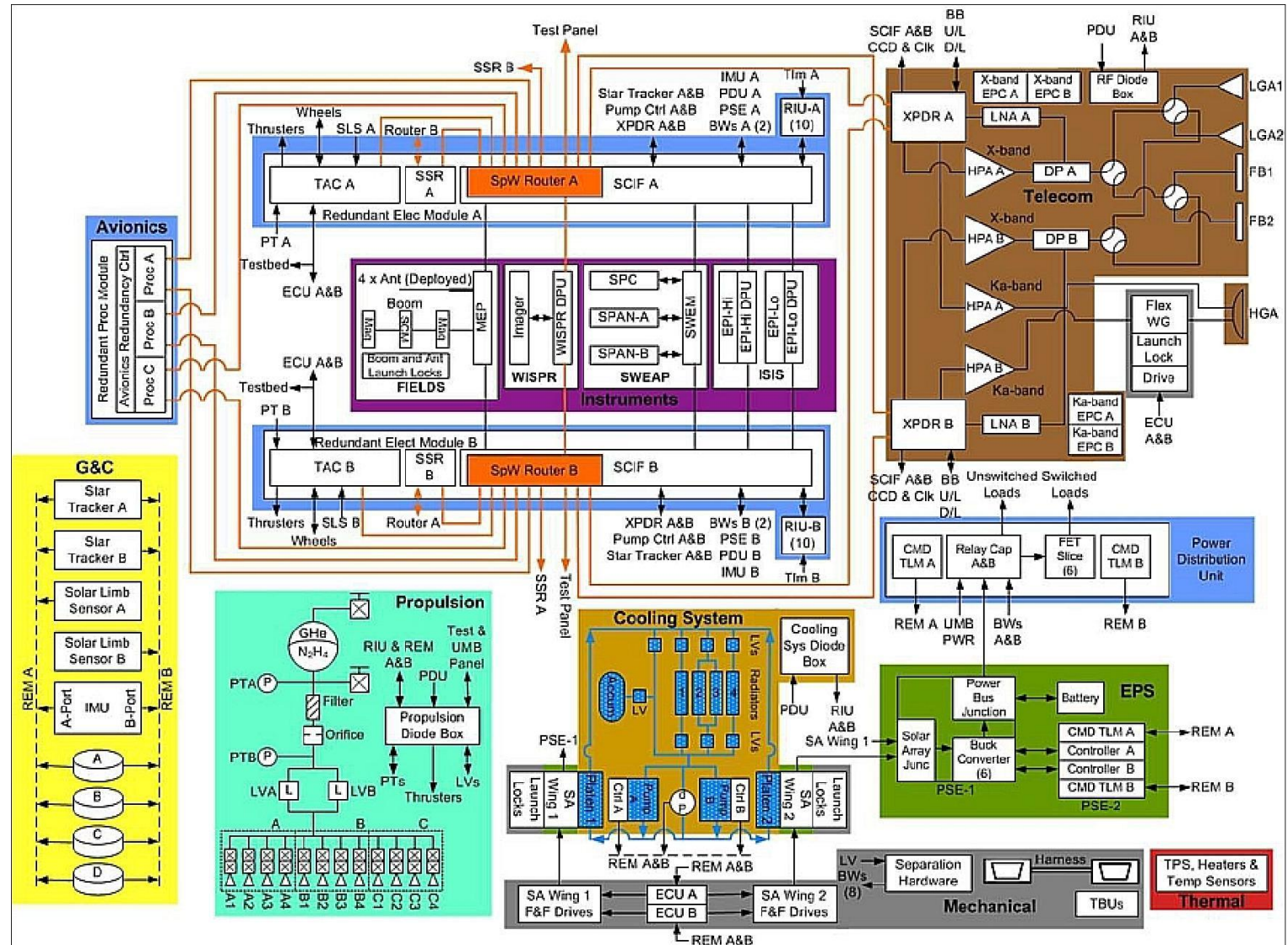
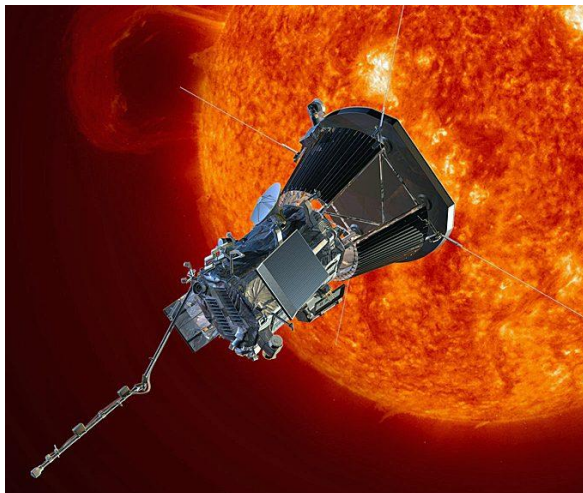
• EMI/EMC

Design Process

Case Studies

• Key EPS design drivers:

- Flagship mission – drives overall redundancy across the system
- Extremely close distance to the Sun – drives precise autonomous solar array control and active solar array cooling
- Severe thermal and radiation environment – drives degradation analysis and environmental testing for solar array and battery



Example: Psyche

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Design Process

Case Studies

• Key EPS design drivers:

- Class B mission – drives overall redundancy across the system
- 3.3 AU from the Sun – drives solar array sizing
- Several hour eclipse periods while in orbital operations around Psyche – drives battery sizing
- Electrical propulsion subsystem – drives high voltage power bus
- Split power subsystem design between JPL and Maxar (formerly SSL), with different heritage architectures – drives use of several power converters, power switching electronics

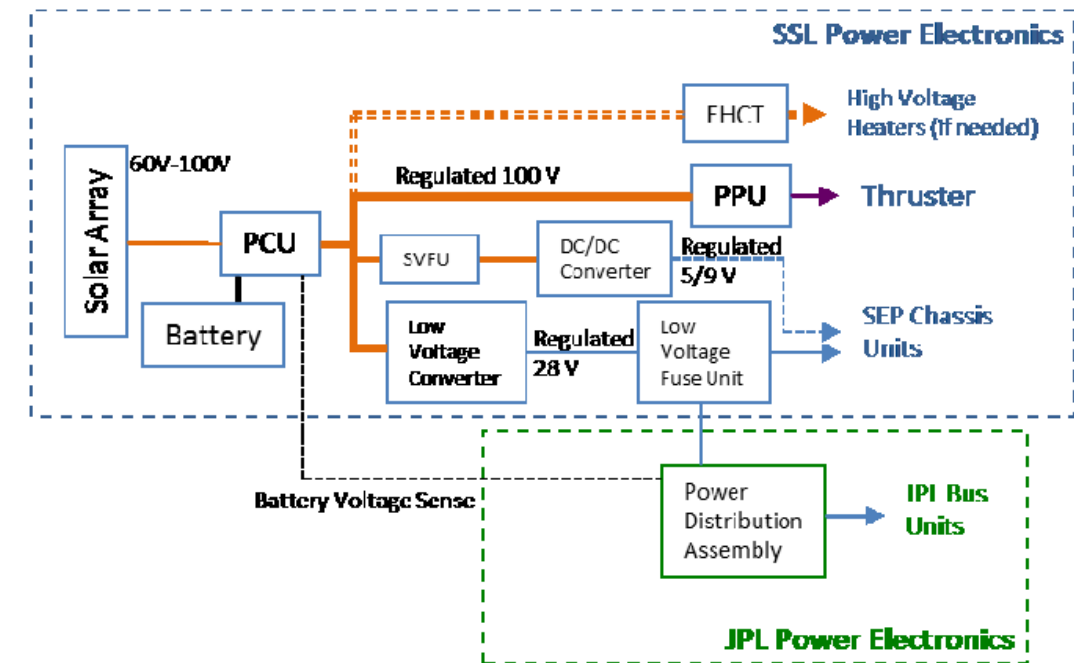


Figure 12. Psyche Spacecraft Power Block Diagram

Power Generation

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Design Process

Case Studies

- **The power generation function on the spacecraft produces electrical energy in order to power all spacecraft subsystems/components, as well as charges the batteries**
- **Typically two forms:**
 - Solar power – uses solar panels to convert sunlight into electrical energy
 - Nuclear power – uses a nuclear power source (e.g., radioisotope thermal electrical generator – RTG) to convert heat from radioactive decay into electrical energy
- **Considerations:**
 - Energy conversion efficiency
 - Solar arrays: ~20 to 30%
 - RTG: ~5 to 10%
 - Degradation
 - Solar arrays degrade from radiation, temperature effects
 - RTGs have much higher longevity
 - Dependence on environment
 - Solar arrays require sunlight; impacted by dust, shadowing
 - RTGs generally independent of environment
 - Total wattage
 - Solar arrays have much higher capability than RTGs
 - Regulatory
 - RTGs subject to nuclear regulations



Power generation

- Solar power
- Nuclear power

Power Generation: Solar Arrays

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Electrical Isolation &

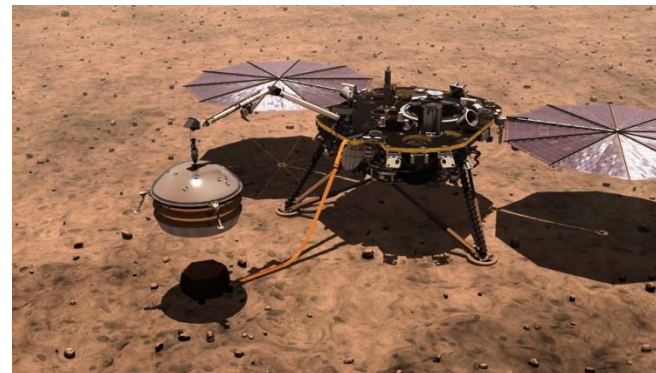
Grounding

- Isolation
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- EMI/EMC

Design Process

Case Studies

- **Solar arrays consist of thousands of interconnected photovoltaic (PV) cells that convert sunlight into electricity**
 - When sunlight hits the solar cells, electrons are excited and generate electric charge
 - This electric charge is then converted to current in order to power the spacecraft devices
- **Solar arrays can be body-mounted or deployable, and deployable solar arrays can be fixed or gimballed**
 - Body mounted: Generally for spinning spacecraft
 - Deployable: Generally for 3-axis stabilized spacecraft or landers
 - Gimballed to maximize direct pointing at the Sun while minimizing Spacecraft turns
- **Design drivers:**
 - Sun distance
 - Radiation environment
 - Thermal environment
 - Micrometeoroids



Power Generation: Solar Arrays

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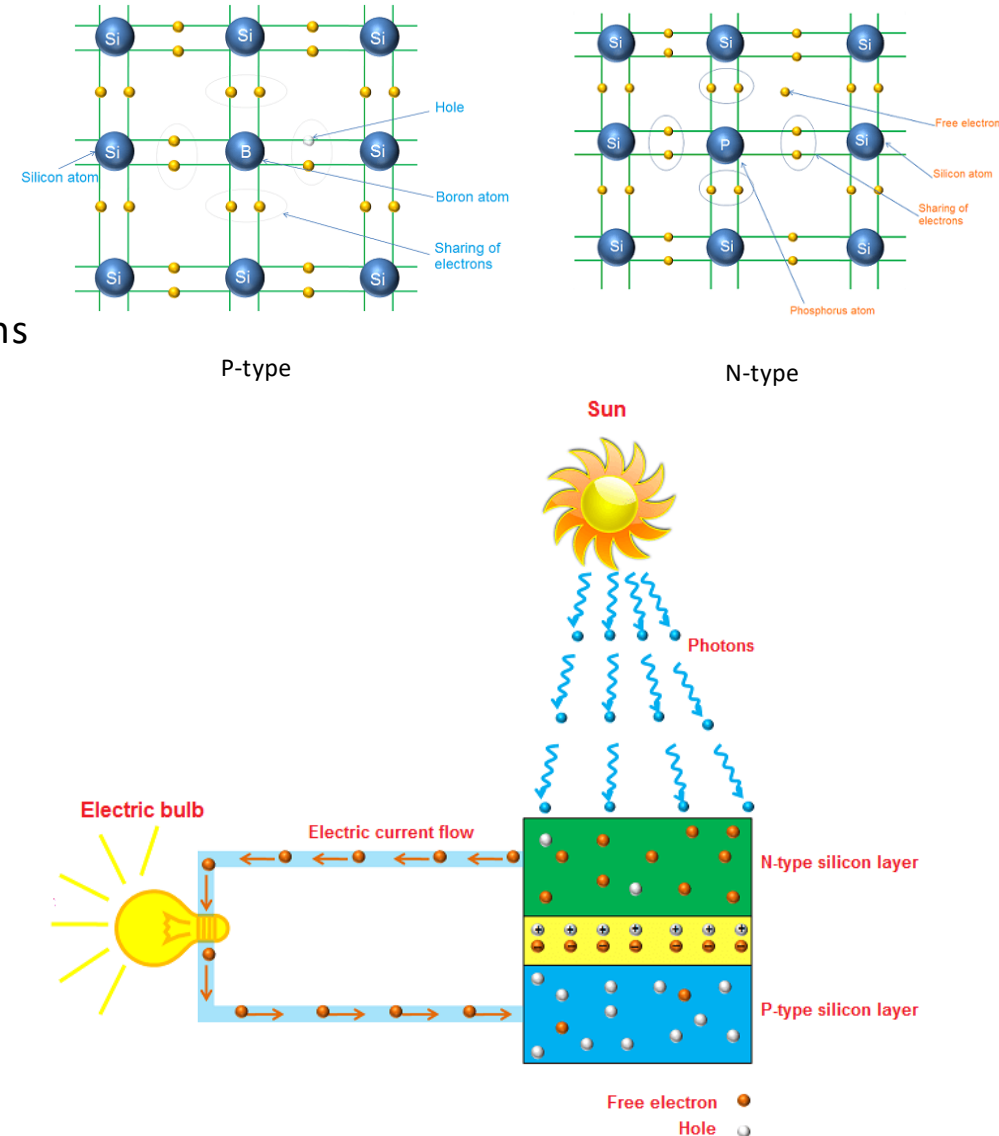
Electrical Isolation & Grounding

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Design Process

Case Studies

- **Solar cells are made from semiconductor materials (e.g., Silicon), consisting of two layers that are “doped” with impurities to create different charge properties**
 - P-type: Positive charge carriers (extra “holes”)
 - N-type: Negative charge carriers (extra electrons)
 - Impurities create excess (n-type) or deficiency (p-type) of electrons
- **When photons from the Sun hit solar cells, it excites and frees electrons, creating electron-hole pairs as electrons move towards holes**
 - This electric charge creates current when connected to a circuit
- **Multi-junction solar arrays use multiple layers of semiconductor material to absorb different wavelengths**
 - Increases efficiency by using more of the solar spectrum
 - Can be sensitive to heat
 - Can be more complex and costly



Power Generation: Solar Arrays

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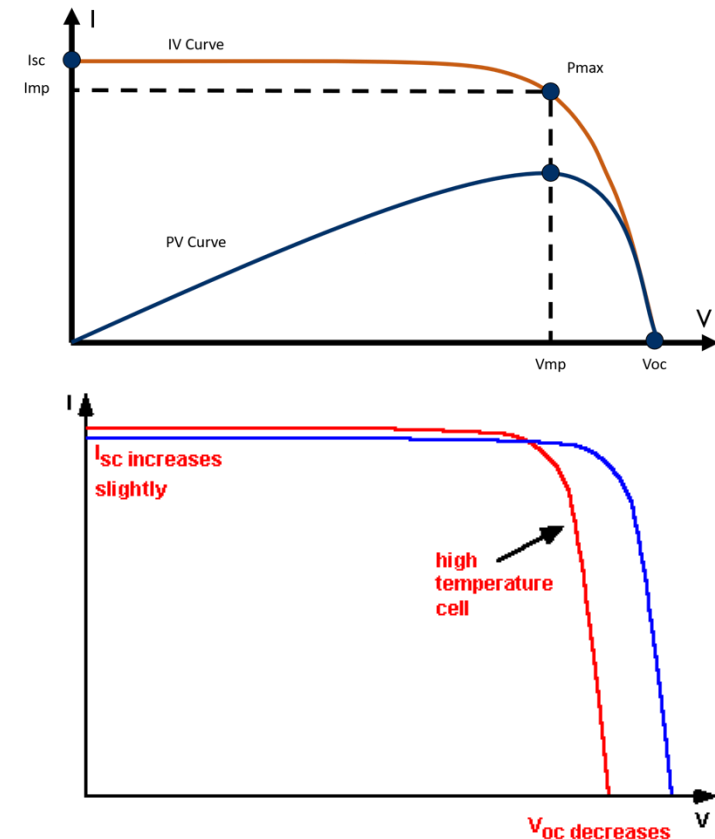
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Design Process

Case Studies

• Solar array current/voltage (I/V) curve

- **X-axis: (Voltage, V):** Output voltage of the solar array
- **Y-axis: (Current, I):** Output current generated by the solar array
- **Short-circuit Current (I_{sc}):** Maximum current generated by the solar array when the output terminals are shorted (i.e., zero resistance) (voltage = 0)
- **Open-circuit Voltage (V_{oc}):** Maximum voltage available from solar array when no load is connected (current = 0)
- **Maximum Power Point (MPP, or P_{max}):** Point on curve where $P = IV$ is maximized; most efficient operating condition for the solar array
- **Usage:**
 - Assess performance of solar panels under different conditions
 - Optimize power system design
 - Size power converters and harness
- **Factors:**
 - Sunlight – more shifts curve up
 - Temp – colder shift curves right
 - Solar panels like cold and sunny!



Analogy:

- Think of a hose, where I = water flow rate and V = water pressure
- I_{sc} : Hose is fully open (no resistance), water flows freely with no pressure
- V_{oc} : Hose is fully closed (infinite resistance), no water flow with pressure at its max
- There is an optimal point where the hose is partially open and the total water “energy” output is at it’s max (MPP)

Power Generation: Solar Arrays

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Design Process

Case Studies

• Solar array output depends on the following factors:

– Solar Irradiance Factor

- Typical solar flux = $1,370 \text{ W/m}^2$ (slight variations with season)
- Scales with distance from Sun by $1/R^2$ (where R is in AU)

– Solar Cell Efficiency (%)

- Typically ~30%, varies by manufacturer/design

– Cosine Loss

- Loss due to the solar incidence angle

– Packing Factor (%)

- Total cell area / total panel area
- Typically ~88%

– Beginning-of-Life (BOL) effects

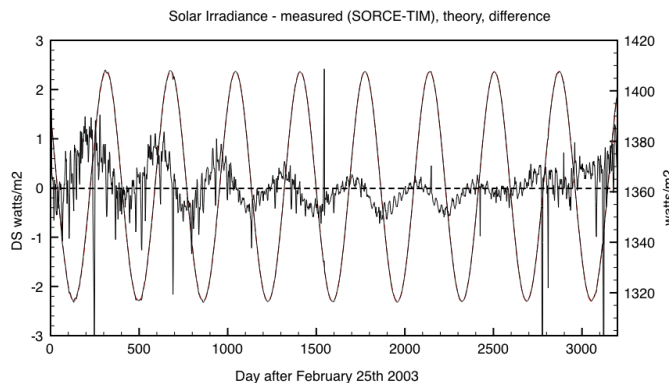
- Workmanship: ~98%
- Wiring losses: ~96%

– End-of-Life (EOL) effects

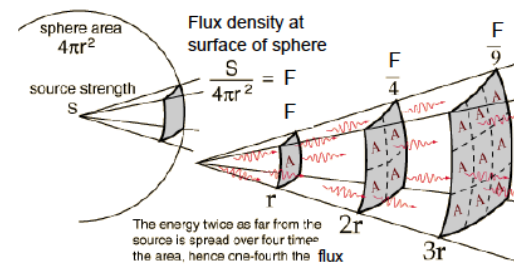
- Radiation: ~94%
- UV: ~98.5%
- Micrometeoroids: ~99.5%

– Solar Array Size (m^2)

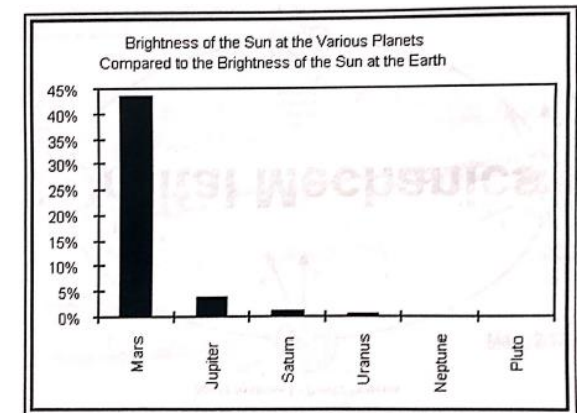
- # of panels x area per panel



Inverse square law



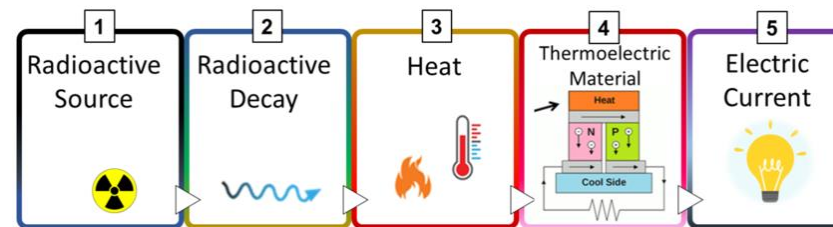
- Irradiance (or flux density) decreases as the square of distance from the source
- Radiance is invariant with distance (note dependence of solid angle on r^2)



$$\text{Array Power (W)} = \text{Solar Flux} \left(\frac{\text{W}}{\text{m}^2} \right) * \left(\frac{1}{R^2(\text{AU})} \right) * \text{Cell Efficiency}(\%) * \text{Cosine Loss} * \text{Packing Factor}(\%) * \text{BOL Effects}(\%) * \text{EOL Effects}(\%) * \text{Array Size (m}^2\text{)}$$

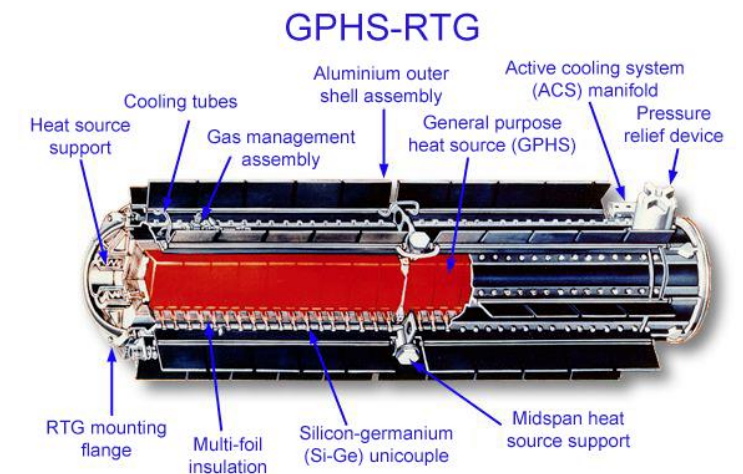
Power Generation: RTGs

- **RTGs generate power through decay of radioactive material (e.g., Plutonium-238)**
 - Decay produces heat, which is converted into electricity using thermoelectric materials that create a voltage due to temperature difference between two sides
- **RTGs are typically used for deep space missions or landers with long eclipse/winter seasons**
 - Power production is on the order of ~200 – 300W
 - Power is steady-state, surge from battery may be needed
- **Example missions**
 - Cassini
 - New Horizons
 - MSL, Mars 2020
- **Design drivers**
 - Nuclear regulations and ground handling
 - Contamination in the case of failures
 - Degradation of thermoelectric material (~1 to 2% per year)
 - Thermal management from heat dissipation (can be harnessed...)



GPHS-RTG

- Used on Cassini, Galileo, and Ulysses
- Weight: 56 kg
- Dimensions: 113 cm length, 43 cm diameter
- Plutonium Load: 10.9 kg
 - 4394 W Thermal Power
- Power:
 - 296 W Electrical at Launch
 - 209 W Electrical EOM



Energy Storage

Energy Storage Overview

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Design Process

Case Studies

- **The energy storage function on the spacecraft stores electrical energy in order to power all spacecraft subsystems/components during periods of time when the solar array or RTG cannot produce sufficient power**
 - Solar array → eclipses for orbiting missions, “overnight” for surface missions
 - RTG → relatively low power generation, battery can supplement
- **Typically a few forms:**
 - Batteries (most common) – converts chemical energy into electrical energy, stores for when needed
 - Can be one-time or rechargeable, used for long duration missions
 - Fuel cells – converts chemical energy into electrical energy, produced continuously
 - Used for continuous power supply when a constant fuel supply can be produced
 - Flywheels – store energy mechanically through a rotating mass
 - Used for short duration applications (e.g., attitude control)
 - Supercapacitors – store energy electrostatically, rapid charge and discharge cycles
 - Used for supplementing high peak loads
- **Considerations:**
 - Energy density
 - Number of charge/discharge cycles
 - Eclipse/off-Sun durations
 - Thermal environment



Energy storage

- Batteries
- Fuel cells
- Flywheels
- Supercapacitors

Energy Storage: Batteries

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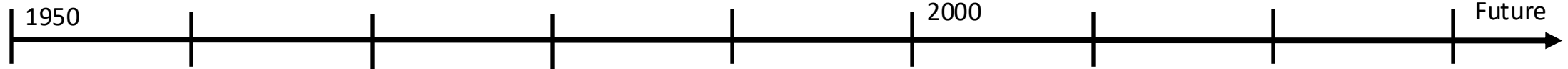
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Design Process

Case Studies



Lead-Acid Batteries: Low-cost, reliable; heavy, low energy density. Used in very early spacecraft (e.g., Voyager, Apollo).

Nickel-Cadmium (NiCd): Improved energy density and thermal properties. Used in early crewed missions (e.g., Gemini).

Nickel-Hydrogen (NiH2): Improved energy density, longer cycle life, and better thermal stability. Used in long-duration missions (e.g., space shuttle).

Nickel-Hydrogen on Hubble



Lithium-Ion (Li-ion): Revolutionized spacecraft power systems - significantly higher energy densities and lighter weights. Used in many modern space systems.

Lithium-Polymer (LiPo): Lightweight alternative to Li-ion. Used in small satellites and cubesats.

Nickel Metal Hydride (NiMH): Improved performance over NiCd with high discharge rates. Good middle ground between cost and performance.

Solid-State Batteries: Higher energy densities and improved safety. Still experiential.

Lithium-Ion Battery (EaglePicher)



Energy Storage: Batteries

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Case Studies

• Terminology and equations:

- **Total capacity of battery, in Ampere-Hours (A-hr):** Amount of current a battery can deliver for a specified time period
 - 40 Amps for 2 hr = 80 A-hr
- **Stored energy of battery, in Watt-Hours (W-hr):** Total amount of energy a battery can provide (depends on voltage)
 - 80 A-hr at 12V = 960 Whr
- **Energy of load, in Watt-Hours (W-hr):** Total amount of energy taken out of battery from spacecraft power loads
- **Depth of Discharge (DOD) (%):** Percent of battery used in discharge
 - 75% DOD means 25% capacity remaining
- **Battery drain (W):** Power required to charge the battery, following discharge (e.g., going in and out of eclipse)
 - Discharge efficiency (%): How effectively a battery can be discharged compared to its total capacity (i.e., how much energy can be used to power loads) – typically ~90%
 - Recharge efficiency: How effectively a battery can be recharged compared to the energy that is put into it (i.e., how much input energy is stored) – typically ~80%
 - Time to recharge: How much time solar array is producing power

$$Capacity (Ahr) = Current (A) * Time (hr)$$

$$Energy_{Battery} (Whr) = Capacity (Ahr) * Voltage (V)$$

$$Energy_{Load} (Whr) = Load (W) * Duration (hr)$$

$$\%DOD = \frac{Energy_{Load} (Whr)}{Energy_{Battery} (Whr)} * 100 = \frac{Capacity_{Used} (Ahr)}{Capacity_{Battery} (Ahr)} * 100$$

Analogy:

- Battery capacity: Like size of a water tank (A-hr = gallons). Tells you how long you can run things before empty (e.g., how long of an eclipse you can survive at a given load).
- Battery stored energy: Amount of “work” water in tank could do (Whr = gallons x pressure). Tells you how much power draw you can provide (e.g., how much load you can support over time).

$$Battery\ Drain\ (W) = \frac{Energy_{Load} (Whr)}{Discharge\ Efficiency\ (\%) * Recharge\ Efficiency\ (\%) * Time\ to\ Recharge\ (hr)}$$

Energy Storage: Other Options

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Design Process

Case Studies

• Fuel cells

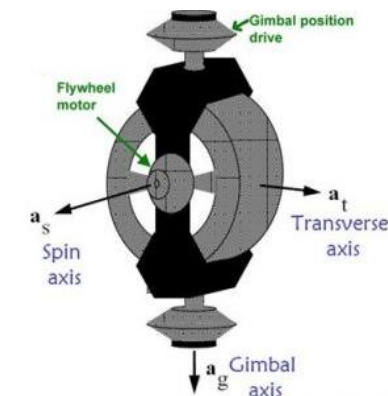
- Converts chemical energy into electrical energy through electrochemical reactions, producing water as the only byproduct
- Uses fuel (typically hydrogen) and oxidizer (typically oxygen)
- Pros: High efficiency, low emissions, able to provide reliable power over extended periods
- Cons: Require a continuous supply of hydrogen and oxygen, which must be carefully managed and stored onboard
- Applications: Apollo, ISS

• Flywheels

- Converts kinetic energy into electrical energy
- Flywheel is spun up to a high rotational speed, which it can then transfer back into an energy storage source such as a battery
- Pros: High efficiency and longevity
- Cons: Can experience mechanical wear, high thermal dissipation
- Applications: Has not seemed to make it beyond studies

• Supercapacitors

- Stores energy electrostatically rather than chemically
- Pros: High power density, rapid charging and discharging capabilities, long cycle life
- Cons: Low energy density, high cost
- Applications: ESA's BepiColombo, Mars Exploration Rovers



Power Control

Power Control: Overview

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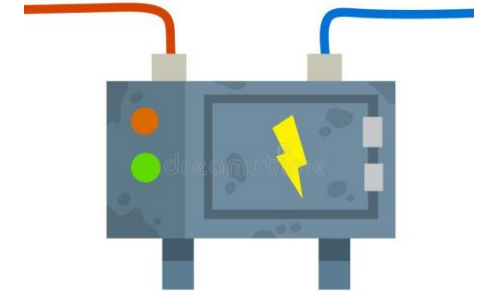
Grounding

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Design Process

Case Studies

- The power control function on the spacecraft converts solar array power into usable power for all spacecraft subsystems/components, provides power switching for all spacecraft subsystems/components, and provides battery charge and discharge functionality
- **Power regulation:**
 - Provides power bus regulation to ensure stable and deterministic operation
 - Provides voltage conversion from solar array and battery to meet the needs of spacecraft components
 - Protects from overload conditions (device drawing more current than it is designed for - for example, due to a short circuit)
- **Power distribution:**
 - Provides power switching for all spacecraft components
 - Provides load isolation to ensure no component can harm the rest of the spacecraft power bus
- **Energy management:**
 - Provides closed-loop battery charging and discharging functionality in hardware and flight software
 - Protects from battery overcharge condition



Power control

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Power Control: Power Regulation

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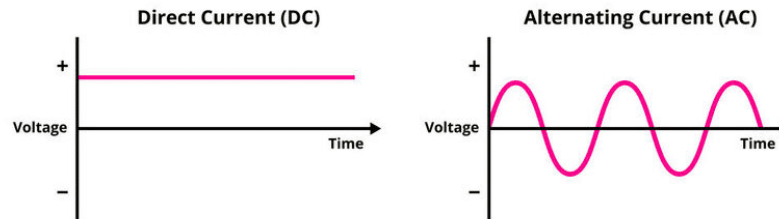
Grounding

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Design Process

Case Studies

- **Most spacecraft run on DC (direct current) power, which provides a steady voltage and current**
 - As opposed to AC (alternating current)



- **Power bus can be regulated or unregulated**
 - Regulated: Output voltage is constant
 - Pros: Provides stability especially for sensitive devices, deterministic
 - Cons: Complex – requires voltage regulators, efficiency losses from voltage regulation
 - Unregulated: Output voltage fluctuates based on input (e.g., solar array power) and load (e.g., power draw from devices)
 - Pros: Simple, highly efficient
 - Cons: Can be damaging for sensitive devices, non-deterministic and challenging to monitor
- **Typical power bus voltage is 28V – 32V**
 - Can be up to 100V depending on system (e.g., EP systems require higher voltage)
 - Instruments sometimes require lower voltages (e.g., 5V, 12V)
 - Power converters can be standalone boxes or embedded in devices (e.g., within flight computer)

Power Control: Power Regulation

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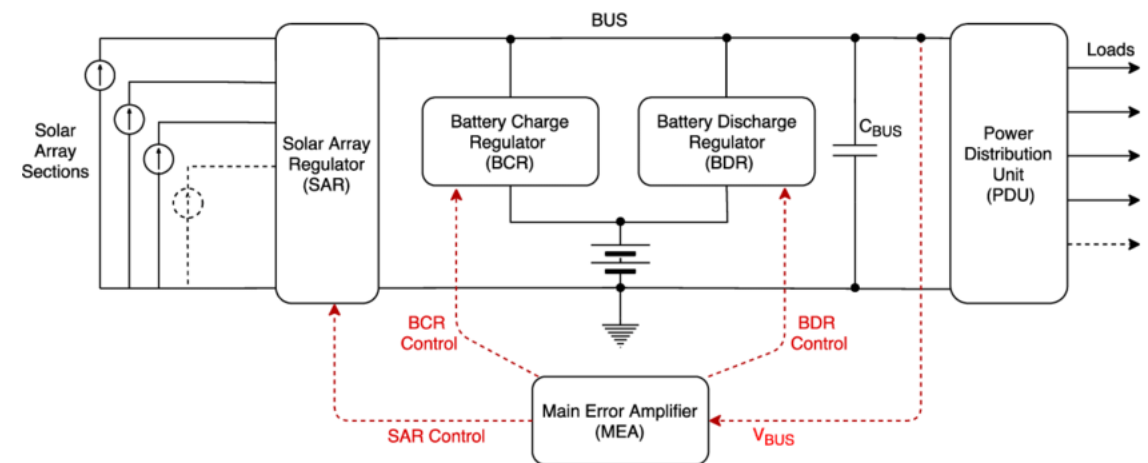
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 - Grounding
 - EMI/EMC
- Design Process
- Case Studies

- **Solar Arrays – Array Power Regulators (or Solar Array Regulators)**
 - Responsible for controlling power output of solar array via monitoring of voltage and current
 - Protects against power overloads
 - Typically sized to handle max expected power draw of solar array
- **Batteries – Battery Charge/Discharge Regulator**
 - Responsible for controlling power output of battery via monitoring of voltage and current
 - Keeps battery charge and discharge current at a safe level
 - Protects against power overloads
 - Typically sized for voltage rating and output power rating
- **Error amplifier**
 - Compares a reference voltage to a feedback voltage from a system and amplifies the difference, known as the error signal
- **All units typically within a spacecraft Power Conditioning Unit (PCU)**



Power Control: Power Regulation

Power

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Power Control

- **Power regulation**
 - Power distribution
 - Energy mgmt
- ### Electrical Isolation & Grounding
- Isolation
 - Grounding
 - EMI/EMC

Design Process

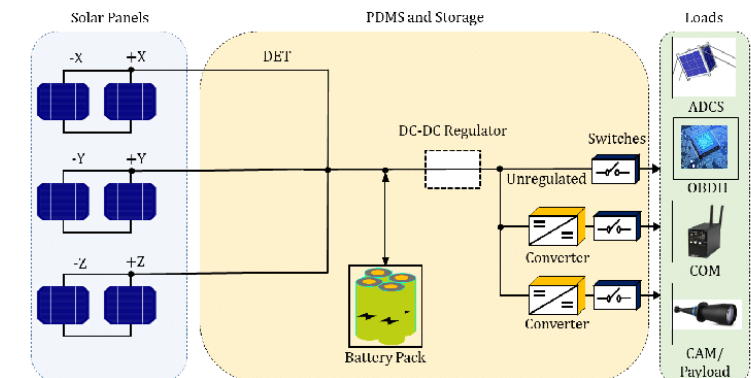
Case Studies

- **Solar array power can be controlled through peak power tracking (PPT) or direct energy transfer (DET)**

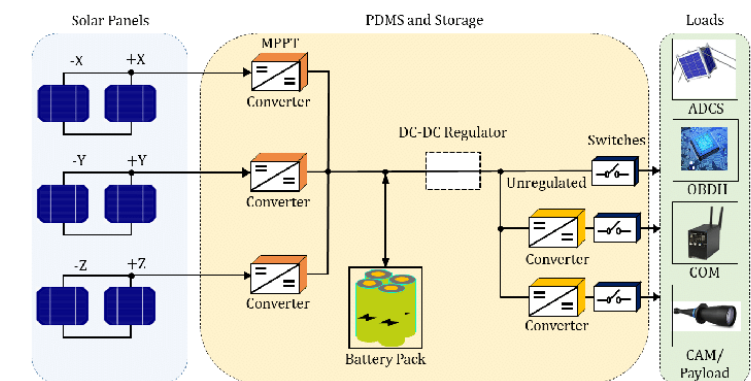
- **Direct energy transfer:** Transfers power directly from solar array to load without intermediary conversion, requires power dissipation for unused power (e.g., shunts, which connect or disconnect portions of solar array)
 - Pros: Highly efficient, simple
 - Cons: Less optimal, variable output
 - Applications: Missions that have fairly continuous power needs
 - Examples: Lunar Reconnaissance Orbiter, Mars Exploration Rovers (DET used for daylight operations, batteries used at night)
- **Peak power tracking:** Uses power converters to dynamically adjust load on solar array to ensure operation at MPP
 - Pros: Maximizes overall power captured
 - Cons: Complex, power conversion results in loss of efficiency
 - Applications: Missions that have variable power needs (e.g., due to changing distance from Sun, dynamic science operations)
 - Examples: Dawn, Juno, Solar Probe

Analogy:

- DET: Riding a bike up a hill fixed in one gear. When the hill gets more or less steep (analogous to sunlight changing, array temp changing), can't adapt pedaling speed (sometimes pedaling too hard or too easy). No gear mechanism to fail (simple) but low efficiency.
- PPT: Riding a bike up a hill with gears. When the hill gets more or less steep, can shift gears to find the right pedaling sweet spot where you produce maximum power most efficiently. Gear mechanism can fail (complex) but high efficiency.

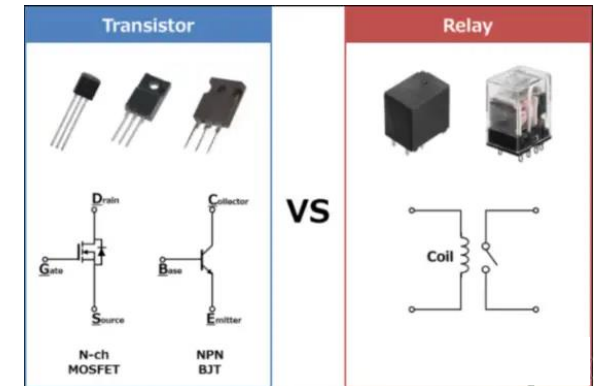


(a)



Power Control: Power Distribution

- **Power distribution across the spacecraft typically handled by a device called the Power Distribution Unit (PDU)**
 - Takes input power from solar array or battery
 - Provides voltage conversion if needed
 - Provides power output to devices via power switches
 - Power switches can be relays or FETs (Field effect transistor)
 - Relays: Use an electromagnet to mechanically open or close a circuit - electrical current flows through a coil, creating a magnetic field that moves an armature to make or break the connection in the circuit (physical device moves)
 - » Pros: Simple, robust
 - » Cons: Slow to open/close, higher mass, can be susceptible to changing position due to mechanical shock/vibration
 - FETs: Use a semiconductor device to control current flow using an electric field – voltage applied to a gate terminal controls flow of current between the source and drain terminals (no moving parts)
 - » Pros: Fast to open/close, lightweight, durable
 - » Cons: More complex, can be susceptible to failure (e.g., due to radiation events)
 - Provides hardware-based protective features such as overvoltage and overcurrent protection (e.g., fusing)
 - Often combined with PCU to form PCDU



Airbus PCDU

Power Control: Energy Management

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- Power distribution

• Energy mgmt

Electrical Isolation &

Grounding

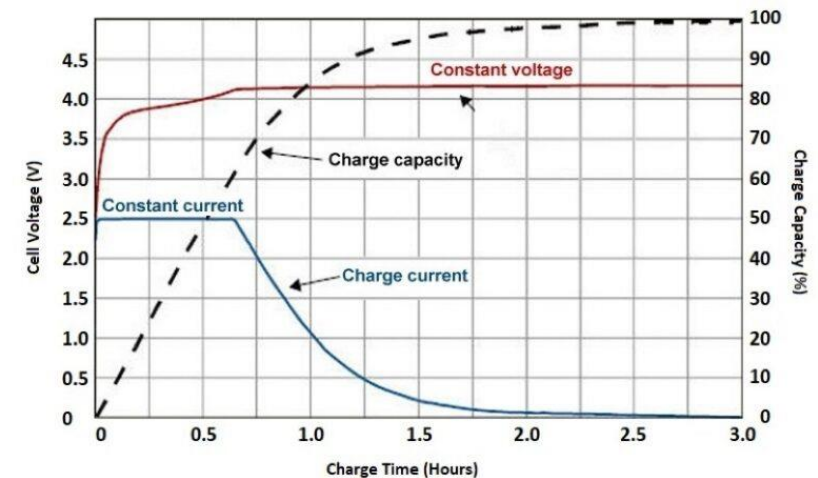
- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies

• Battery charging process:

- Energy produced by solar array is conditioned by PCU
- Electronics within PCU, as well as flight software, regulate charging process, including:
 - Monitoring battery voltage and/or state of charge to determine when to start/stop battery charging
 - Adjusting charging rate by monitoring voltage and current
 - Monitoring temperature to ensure battery does not overheat
- Phases:
 - Bulk charge: Battery is charged at a constant current until it reaches a set voltage
 - Absorption charge: Battery voltage is held constant while current gradually decreases as battery nears full charge
 - Trickle charge: Battery is charged at a low, constant current to maintain its charge without overcharging
- Safety considerations:
 - Overcharge protection: Prevent battery from exceeding safe voltage condition
 - Thermal management: Prevent battery from overheating, thermal runaway
- Battery is typically maintained at a lower voltage than capacity, for lifetime considerations

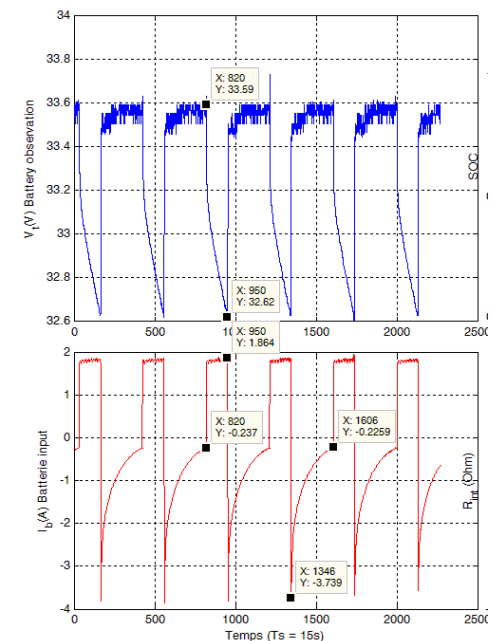


Note: Battery voltage and state of charge (% capacity) are not the same. Both represent how "charged" a battery is, but battery voltage depends on temperature, load, cell resistance, etc.

Charge/discharge cycles of a spacecraft in LEO

Constant current input while in Sunlight, battery charges and voltage increases until full capacity

Zero current in eclipse, battery discharges and voltage decreases



Electrical Isolation and Grounding

Electrical Isolation

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Electrical Isolation & Grounding

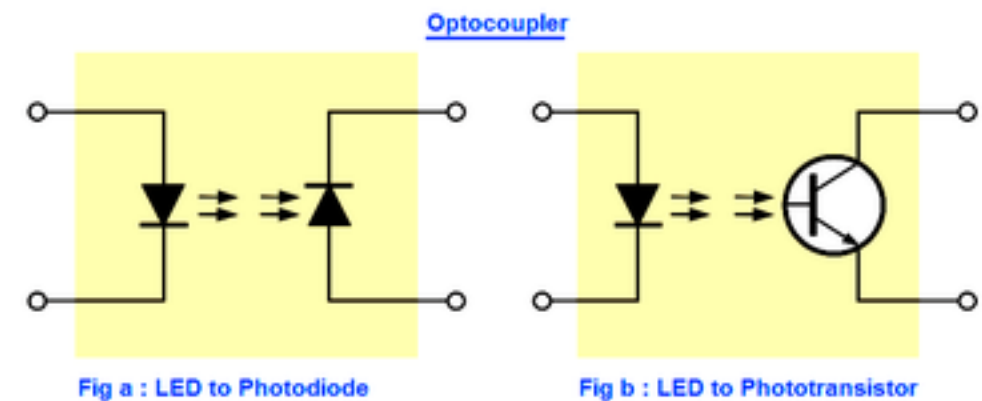
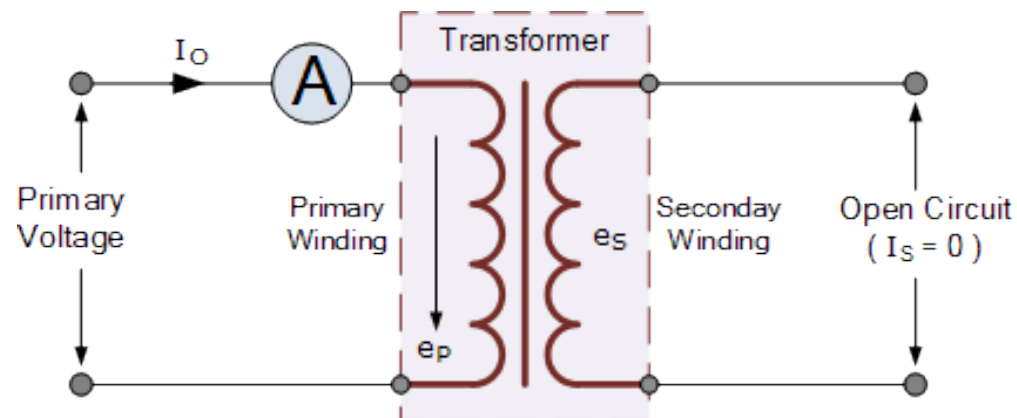
- **Isolation**
- Grounding
- EMI/EMC

Design Process

Case Studies

• Electrical isolation on spacecraft prevents unwanted electrical connections and/or interference

- Critical reliability function; protects from high voltages, prevents ground loops (multiple paths to ground), and minimizes noise and interference between circuits
- Methods:
 - **Physical Separation:** Keep high-voltage systems physically separate from low-voltage systems to prevent electrical contact
 - **Isolation Barriers:** Implement coatings or enclosures to prevent electrical contact
 - **Transformers:** Isolate circuits by transferring energy through electromagnetic induction (magnetic fields)
 - **Optocouplers:** Isolate circuits by transferring energy through light



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Electrical Isolation & Grounding

- Isolation
- Grounding**
- EMI/EMC

Design Process

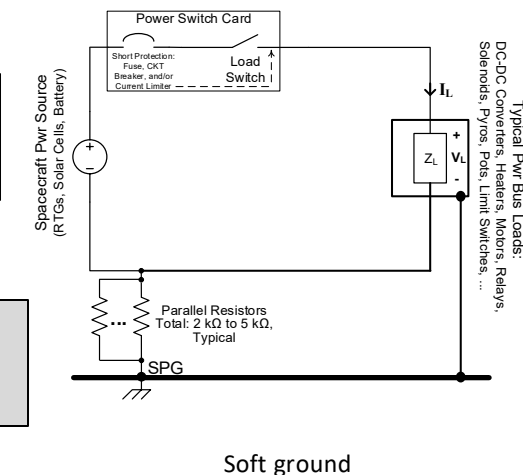
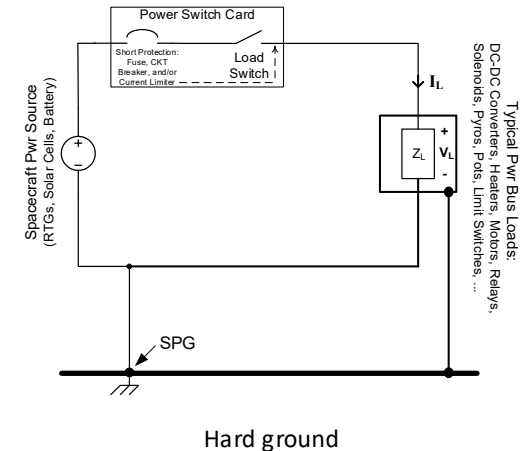
Case Studies

- Grounding on spacecraft establishes a common reference point for electrical circuits and establishes a path for current to safely dissipate**
 - Critical reliability function; protects against electrical shock, minimizes electromagnetic interference (EMI), provides a path for fault currents to dissipate
 - Single Point Ground (SPG):** Establishes a common ground point for all systems to prevent ground loops and minimize potential differences (typically at Spacecraft structure)
 - Grounding schemes:**
 - Hard Ground:** Uses a low-resistance connection to SPG
 - Pros: Stable and reliable reference, safely dissipates fault currents
 - Cons: Inflexible, susceptible to noise and interference due to ground loops
 - Applications: Power distribution systems, safety systems
 - Soft Ground:** Uses a high-resistance connection to SPG
 - Pros: Flexible, minimizes noise and interference
 - Cons: Not very effective in dissipating fault currents
 - Applications: Sensitive circuits that require isolation from high voltage
 - Floating Ground:** Completely isolated from SPG
 - Isolation achieved using transformers, optocouplers, etc
 - Pros: Minimizes noise and interference, even more than soft ground
 - Cons: Can vary significantly, not very effective in dissipating fault currents
 - Applications: Sensitive circuits that require isolation from high voltage

Analogy: Connecting a boat to a dock with a rigid rod – very strong, but very susceptible to waves

Analogy: Connecting a boat to a dock with a bungee cord – less strong, but less susceptible to waves

Analogy: A boat floating free of the dock – lowest strength, but least susceptible to waves



Power

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Electrical Isolation &

Grounding

- Isolation
- Grounding
- EMI/EMC

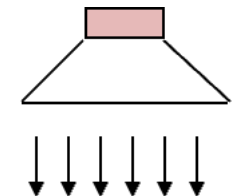
Design Process

Case Studies

- **Electromagnetic Interference (EMI): Disturbance caused by electromagnetic radiation to electronic devices**
 - Types of EMI:
 - Conducted: Occurs through conductive paths (direct connection)
 - Radiated: Occurs through propagation through space (indirect connection)
 - Design sources:
 - Internal: Motors, power supplies, digital circuits, devices turning on and off
 - External: Solar flares, cosmic radiation, other spacecraft
- **Electromagnetic Compatibility (EMC): Ability of a device to function without causing interference to or interfering with other devices**
 - Types of EMC:
 - Emissions: Limiting amount of EMI that a device generates (device is “source”)
 - Susceptibility: Ensuring device is immune to EMI from both internal and external sources (device is “victim”)
 - Design considerations:
 - Physical separation: Keeping devices physically far apart from each other
 - Shielding: Implementing physical barriers using conductive or magnetic materials, separate wiring, etc
 - Filtering: Blocking unwanted frequencies from entering or leaving signal paths
 - Grounding: Providing a return path for electromagnetic energy to ground loops and minimize noise

EMI/EMC have a noise source, a coupling mechanism, and a victim circuit

- Motors
- Power supplies
- Radiation



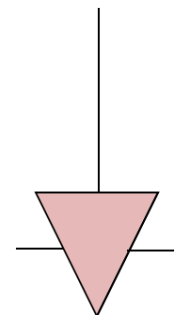
Noise Source

- Shared power lines
- Close proximity
- Ground noise



Coupling Mechanism

- Poor shielding
- No filters
- Poor grounding



Victim Circuit

Power Subsystem Design Process

Power Subsystem Design Process

Overview

Power

Introduction

- Overview
- Examples

Power Generation

- Solar arrays
- RTGs

Energy Storage

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Power Control

- Power regulation
- Power distribution

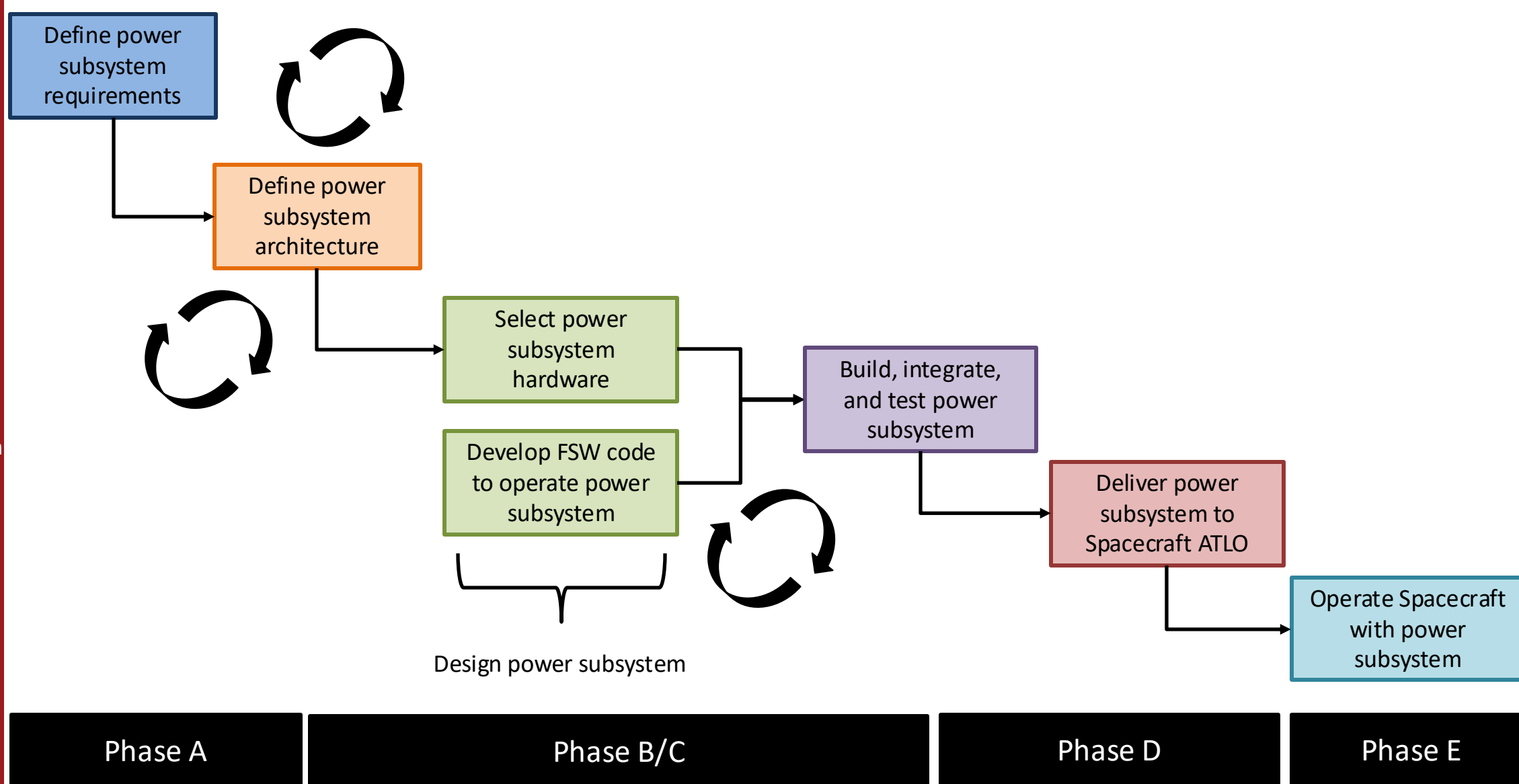
Electrical Isolation &

Grounding

- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Requirements

Power

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Power Generation

- Solar arrays
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Energy Storage

- Batteries
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Power Control

- Power regulation
- Power distribution
- Energy mgmt

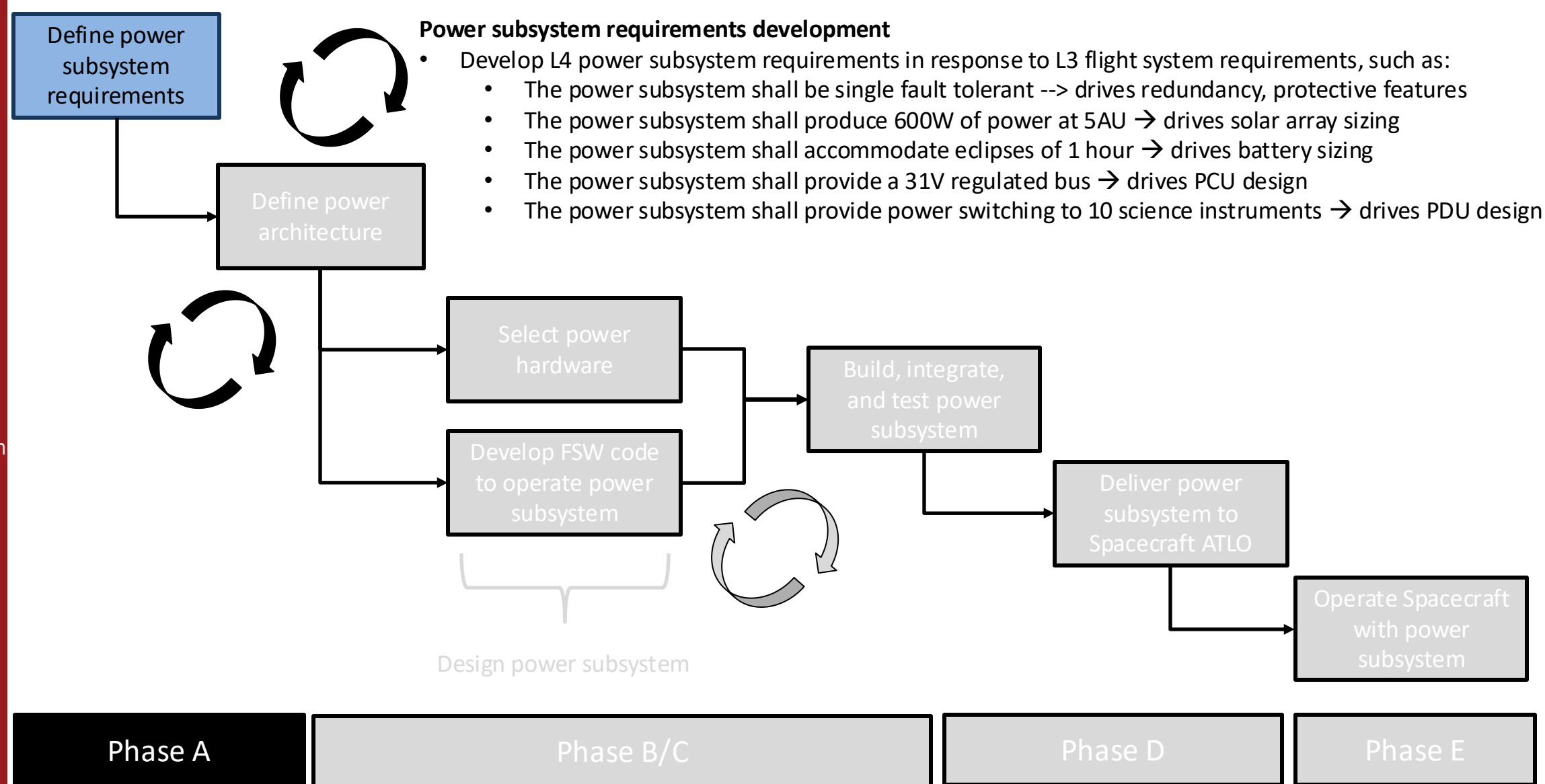
Electrical Isolation &

Grounding

- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Architecture

Power

Introduction

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Power Generation

- Solar arrays
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Energy Storage

- Batteries
- Other

Power Control

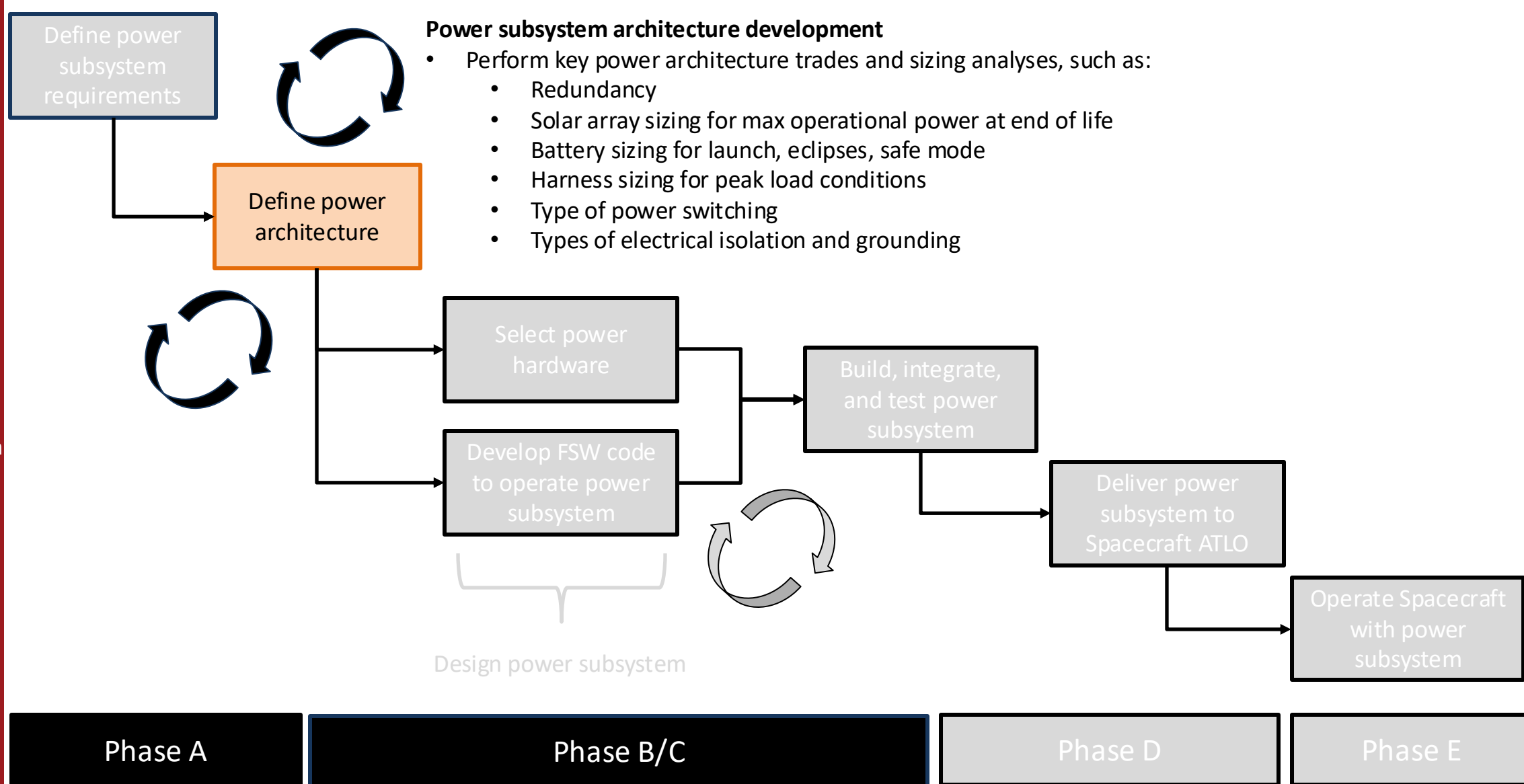
- Power regulation
- Power distribution
- Energy mgmt

Electrical Isolation & Grounding

- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Design

Power

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- Solar arrays
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Energy Storage

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Power Control

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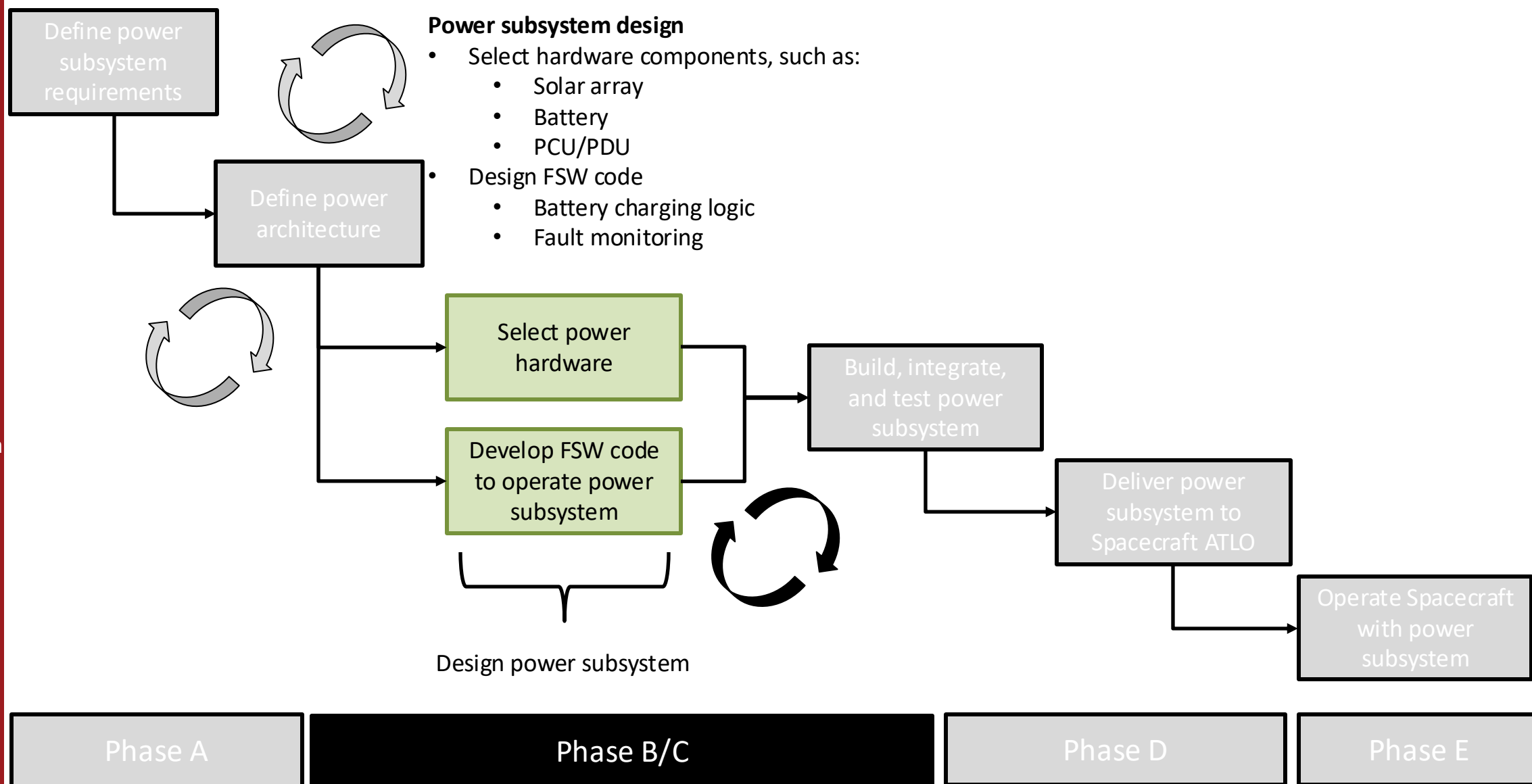
Electrical Isolation &

Grounding

- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Build and Test

Power

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- Solar arrays
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Energy Storage

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Power Control

- Power regulation
- Power distribution

Electrical Isolation &

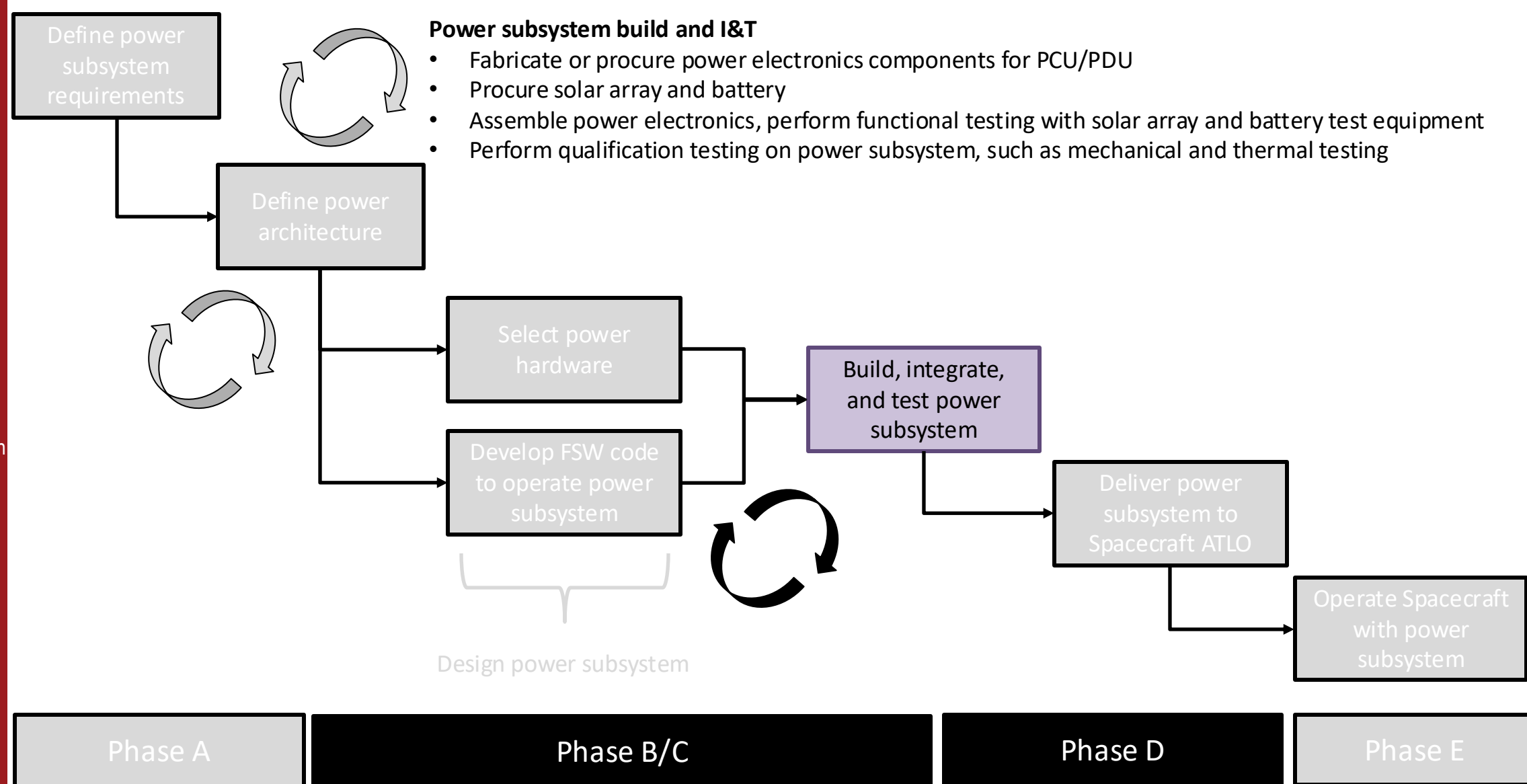
Grounding

- Isolation
- Grounding

- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Deliver

Power

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Power Generation

- Solar arrays
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Energy Storage

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- Other

Power Control

- Power regulation
- Power distribution
- Energy mgmt

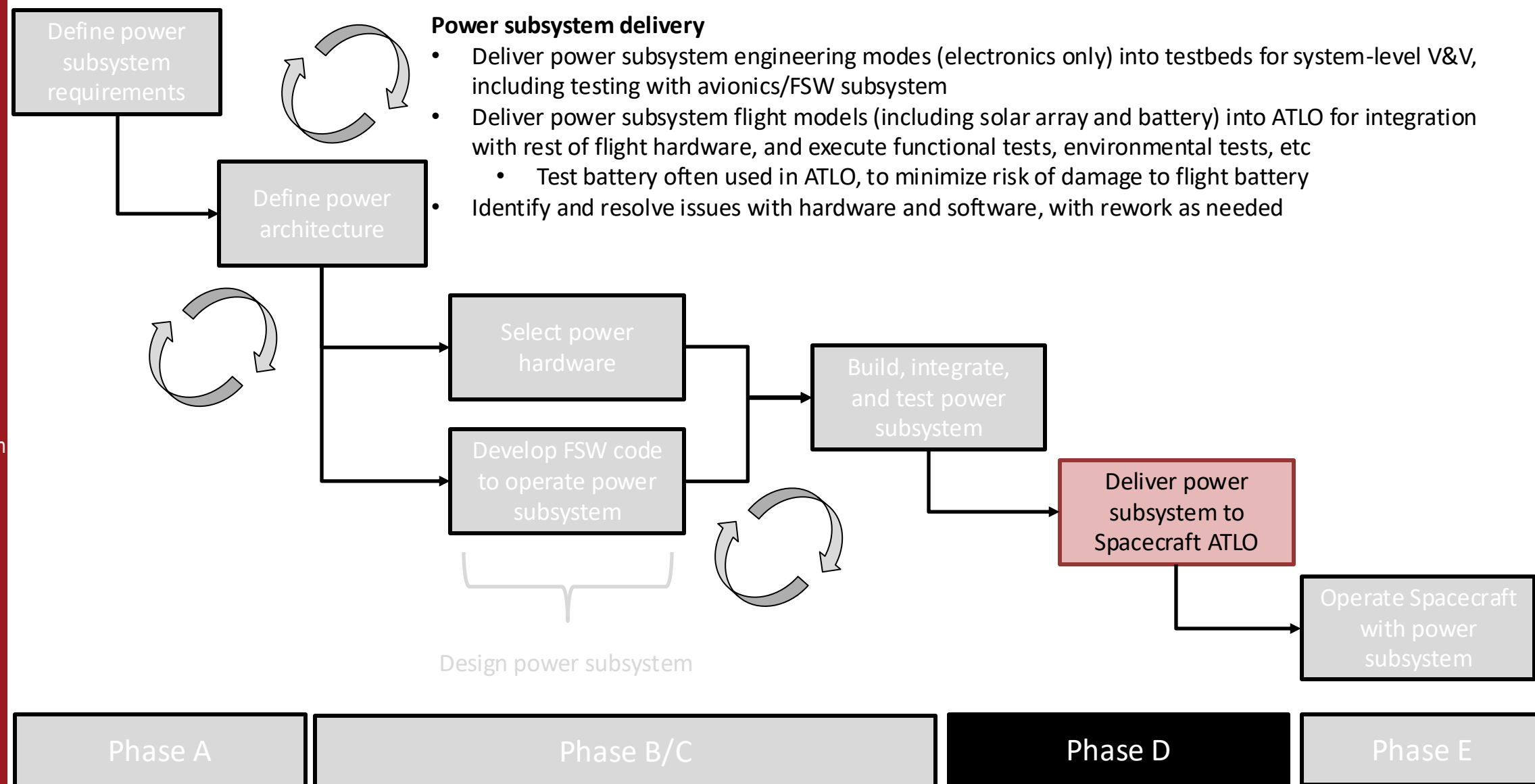
Electrical Isolation &

Grounding

- Isolation
- Grounding
- EMI/EMC

Design Process

Case Studies



Power Subsystem Design Process

Operate

Power

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Power Control

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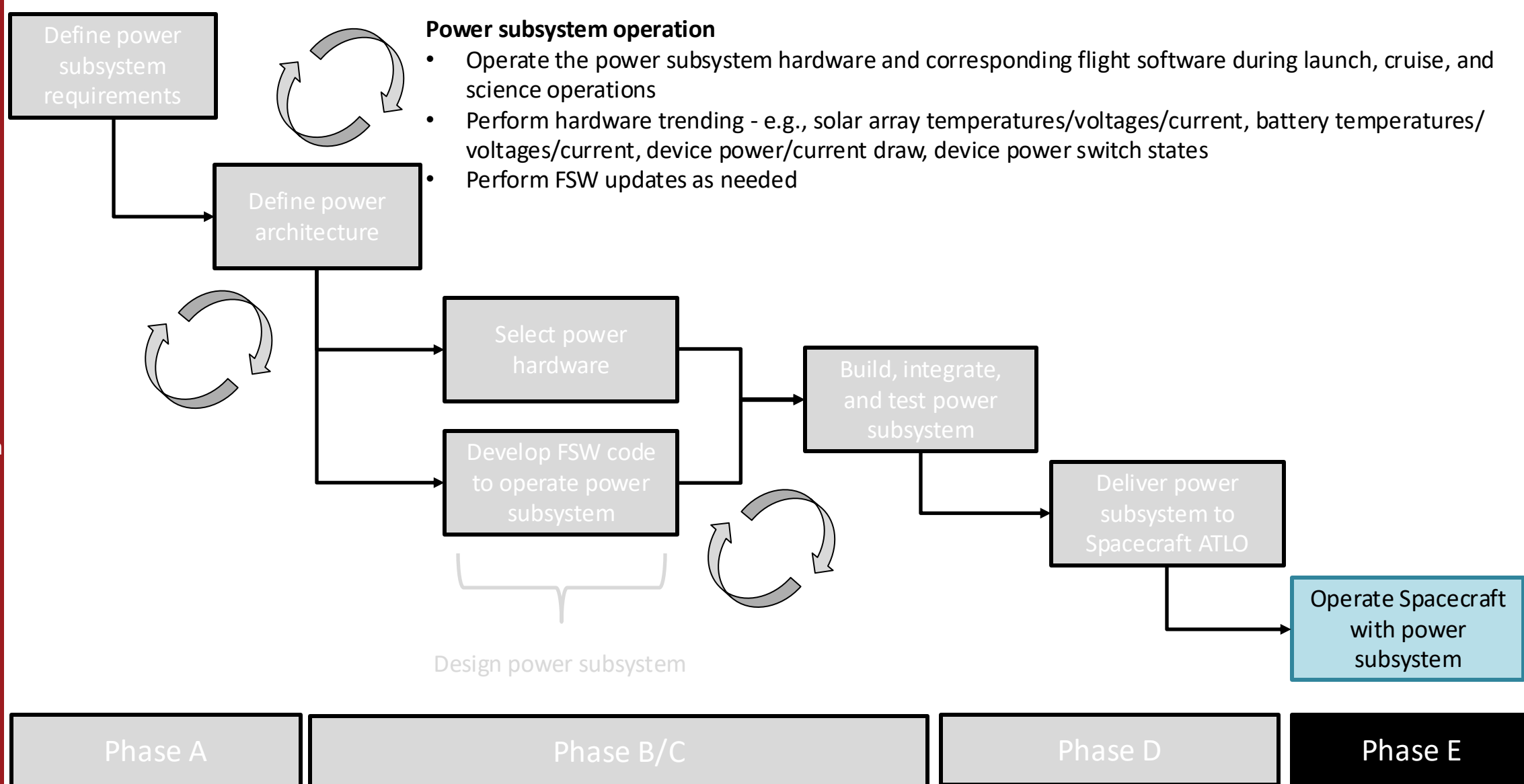
Electrical Isolation &

Grounding

- Isolation
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- EMI/EMC

Design Process

Case Studies



Case Studies

Case Study: Europa Clipper

Power

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Isolation

Grounding

EMI/EMC

Design Process

Case Studies

- **Europa Clipper has two battery inhibits, to ensure battery is safely connected or disconnected**
 - Battery Arm Plugs: When installed, complete the circuit and allow the batteries onto the power bus
 - Battery Discharge Interrupt Switches: Can be held OPEN if continuously driven by ground support equipment (GSE) to power off spacecraft
 - If the bus voltage reaches an extreme undervoltage condition, the switches autonomously OPEN
 - Previous missions, (like M2020/MSL) used latching relays that could be pulsed open or close without continuous power supply
- **Europa Clipper had installed the Battery Arm Plugs in order to encapsulate and stack onto the Falcon Heavy launch vehicle**
 - Encapsulation of Europa Clipper into the launch fairing prevented access to remove the Battery Arm Plugs
 - Once Battery Arm Plugs are installed, Europa Clipper was powered ON and required 24/7 monitoring from the ATLO team for many days prior to launch
 - Members of Europa Clipper ATLO team were required to monitor the GSE throughout Hurricane Milton in order to ensure that the Battery Discharge Interrupt Switches were held OPEN keeping Europa Clipper powered OFF
 - Had latching relays been used, the relays could have been pulsed OPEN and the ATLO team could have safely evacuated the area until the Hurricane passed

